Pharmaceutical Report

Group 22

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Executive Summary

This project investigated how time and district-level socioeconomic deprivation were associated with the dispensing of two medications used in the treatment of dementia, donepezil hydrochloride and rivastigmine, across New Zealand health districts from 2020 to 2024. Pharmaceutical dispensing data were combined with the New Zealand Index of Socioeconomic Deprivation and annual estimates of the population aged 65 years and over. The pharmaceutical data were restricted to dispensing records, geographic names were harmonised across datasets, and population-adjusted dispensing rates were calculated to account for differences in the size of the older population between districts. Exploratory analysis showed that national dispensing counts for both medications increased over the study period, although substantial differences were observed between districts and between the two medications. Older population size explained a considerable amount of the variation in raw dispensing counts, supporting the use of population-adjusted rates. The relationship between socioeconomic deprivation and dispensing also appeared to differ by medication and over time. A series of Poisson, negative binomial and mixed-effects models were compared. The preferred model was a negative binomial mixed-effects model that included year, average district deprivation, medication type and their interactions, adjusted for the population aged 65 years and over, with a random intercept for district. At the average level of deprivation, the model estimated that the dispensing rate of donepezil hydrochloride increased by approximately 3.0% per year, while the dispensing rate of rivastigmine decreased by approximately 3.3% per year. The association with deprivation

also differed between the medications and changed over time, indicating that the effects of time and socioeconomic deprivation cannot be considered independently of medication type.

These findings describe associations rather than causal relationships. The analysis is limited by the use of district-level rather than individual-level data, the relatively small number of health districts, the repeated observation of the same districts, and the lack of annually updated deprivation measurements. In addition, pharmaceutical dispensing records indicate that a medication was dispensed, but do not confirm that it was taken by the patient. The results therefore provide evidence of geographic, temporal and socioeconomic variation in dementia-medication dispensing while highlighting the need for caution when interpreting the causes of these differences.

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Background

Medicines are an important part of healthcare, and examining pharmaceutical dispensing data can help us understand how medicine use varies across New Zealand, between geographic areas and over time. Dispensing patterns may be influenced by differences in population size, age structure, socioeconomic conditions and healthcare needs. This project uses pharmaceutical dispensing data to explore these patterns, with a particular focus on medicines associated with dementia.

Dementia describes a group of conditions that cause a progressive decline in cognitive functions such as memory, thinking and the ability to carry out everyday activities. It is more common among older people and can create substantial healthcare and support needs. Understanding patterns in dementia-related pharmaceutical dispensing is therefore important, particularly as differences may exist between geographic areas and population groups.

The group's analysis focuses on Donepezil hydrochloride and Rivastigmine, two medicines used in the management of symptoms associated with dementia. We examine how dispensing of these medicines changed between 2020 and 2024 and how patterns differed between New Zealand health districts. However, raw dispensing counts alone may not provide a fair comparison between districts. For example, an area with a larger population aged 65 years and over may naturally have more dementia-related medicine dispensings. Population and socioeconomic deprivation information were therefore incorporated to provide additional context when examining geographic differences.

The pharmaceutical data were obtained from Te Whatu Ora – Health New Zealand and contain aggregated information about medicines dispensed through community pharmacies in New Zealand. The initial exploratory analysis examines general pharmaceutical patterns, including commonly dispensed medicines, changes over time, geographic variation and therapeutic groups, before focusing more closely on Donepezil and Rivastigmine.

Data

Three main sources of data were used: pharmaceutical dispensing data, socioeconomic deprivation data, and population data. The pharmaceutical dispensing data were obtained from Health New Zealand | Te Whatu Ora's Pharmaceutical Data Web Tool. The socioeconomic deprivation data and population estimates were obtained from [https://explore.data.stats.govt.nz/vis?pg=0&bp=true&snb=3&tm=Population%20estimates%20health%20district&hc%5BArea%5D=&df%5Bds%5D=ds-nsiws-disseminate&df%5Bid%5D=POPES_SUB_002&df%5Bag%5D=STATSNZ&df%5Bvs%5D=1.0&dq=2020%2B2021%2B2022%2B2023%2B2024%2B2025.3.4%2B14%2B15%2B16%2B17%2B18%2B19.100%2B101%2B102%2B103%2B200%2B201%2B202%2B203%2B204%2B300%2B301%2B302%2B303%2B304%2B400%2B401%2B402%2B403%2B404%2B900&to%5BTIME%5D=false&isAvailabilityDisable false]. These datasets have different levels of granularity and were integrated where required for the dementia analysis.

The pharmaceutical data were provided in several CSV files, including Data_ByChemical.csv, Data_ByTG2.csv, Datya_ByTG3.csv and PharmaceuticalsLookup.csv. The chemical-level dataset was the main dataset used for analysing individual medicines. Each row represents an aggregated combination of characteristics such as pharmaceutical chemical, health district, year and dispensing type, rather than an individual patient. The TG2 and TG3 datasets contain more aggregated therapeutic-group and subgroup information, while the lookup table connects individual pharmaceutical chemicals to their corresponding therapeutic classifications.

The socioeconomic deprivation data provide information on deprivation across geographic areas. For the analysis, these data were used to calculate a population-weighted deprivation measure for each health district, providing district-level socioeconomic context. The population data contain population estimates by geographic area, year and age group. For the dementia analysis, these were aggregated to obtain estimates of the population aged 65 years and over for each district and year. This allowed dementia-related dispensing patterns to be considered relative to the size of the older population rather than comparing districts using raw dispensing counts alone.

Variable	R data type	Description
Chemical	Character	Name of the pharmaceutical chemical
ChemID	Character	Identifier associated with the pharmaceutical
chemical		
District	Character	Health district or New Zealand national aggregate
YearDisp	Numerical	Year in which the dispensing was recorded

Variable	R data type	Description
NumDisps	Numerical after cleaning	Number of pharmaceutical dispensings
NumPpl	Numerical after cleaning	Number of people associated with the dispensing record

Figure 1. Main variables in the chemical-level pharmaceutical dataset

Although variables such as Chemical, District and Type are stored as character variables in R, they can be treated statistically as categorical variables. NumDisps and NumPpl were initially stored as character variables because some observations contain the value <6. After treatment of these suppressed observations, they were converted to numerical variables.

Data quality and preparation

Before the main analysis, the pharmaceutical data were checked for missing values, duplicate records and variable structure. Records were then filtered to include Type = “Dispensings”, excluding initial dispensing records before suppression was assessed. After this filtering, 4127 records had suppressed NumDisps values and 10147 records had suppressed NumPpl values.

Suppression occurs when a small count is reported as <6 rather than providing its exact value. For numerical analysis, these values were replaced with 3 for both NumDisps and NumPpl, allowing the observations to remain in the analysis rather than being removed. However, the actual suppressed values are unknown, so imputing a value of 3 introduces a small amount of uncertainty into calculated totals, rates and subsequent analyses.

Integration of the three data sources also required attention to the join keys. District names were not completely consistent between datasets because of differences in naming conventions and the inclusion of Māori macrons. For example, Tairawhiti and Waitemata in the pharmaceutical data were standardised to Tairawhiti and Waitemata. Capital & Coast and Hutt Valley were also combined as Capital, Coast and Hutt Valley where required to match the geographic structure of the other data. Standardising these names was necessary because inconsistent join keys could result in records for the same geographic area failing to match.

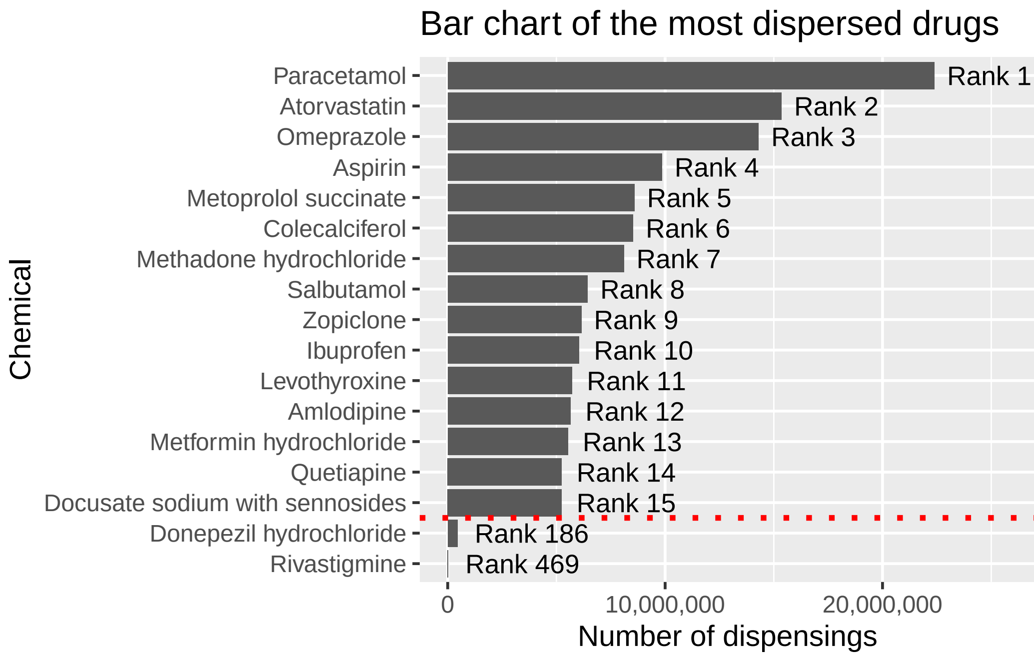
After standardisation, pharmaceutical and deprivation information could be linked by district, while population information was linked using district and year. These integrated data provide the basis for examining whether differences in Donepezil and Rivastigmine dispensing across New Zealand are associated with geographic, demographic and socioeconomic characteristics.

Ethics, Privacy and Security

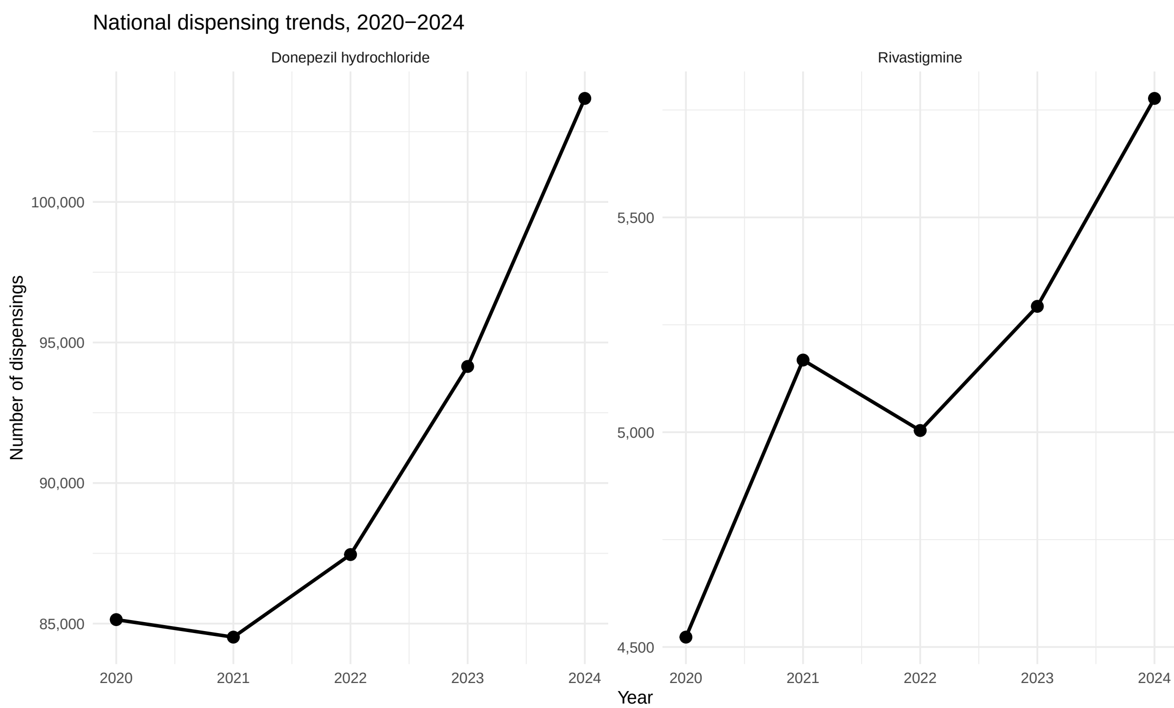
- Ethical considerations:
  - Any comparison of groups being done in this project shouldn't cause have any implications about financial status between said groups. For example, Māori people take less medication than Pākehā because they can't afford them.
  - Given the nature of the data collection, no causal causal conclusions can be drawn. The analysis will only be able to create commentary about the trends and relationships between the variables.
- Privacy considerations:
  - We should not be removing the suppression layer from the count of medication distribution. While there is an oversight in how this data is provided that allows users to do this, and there is code demonstrating that this is possible in this project, the values were randomly adjusted so that for the example instance doesn't have the true count revealed.
  - For all analysis, the suppressed values will be imputed. Currently the method of replacing these with the median of the suppression range.

Exploratory Data Analysis

The graph below shows an unequal distribution of dispensing of medications of the 15 most widely distributed chemicals. Paracetamol is by far the most dispensed, with over double the amount of dispensings compared to the aspirin, the chemical in 4th place. The second and third most widely distributed medications are atorvastatin and omeprazole. Overall, the distribution has a long tail. Donepezil hydrochloride and Rivastigmine aren't highly distributed enough to make the top 15, with the former ranking 186th across the 5 years of data, and the latter ranking 469th over the same period.



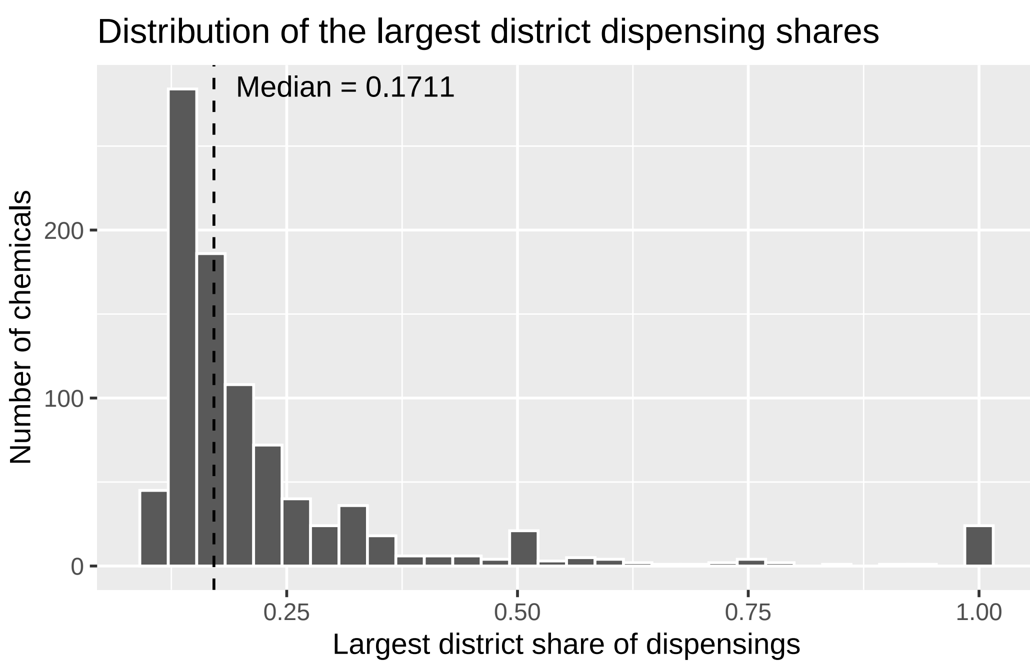
Looking at the line charts below shows the dispensing count for the two primary dementia medications annually from 2020 to 2024. Both show an upward trend of distributions over time.



The table below shows the same information in table format, broken down by count and ranks of the two dementia medications of interest. There has been an upwards trend in dispensing for both of these medications, while rivastigmine has had its rank reduced in recent years. This is because while the dispensing of this medication has increased in these years, not as much as other medication that had a lower dispensing count that rivastigmine in 2021.

Year	DH count	R count	DH rank	R rank
2020	85142	4523	188	466
2021	84519	5168	189	454
2022	87456	5004	192	457
2023	94148	5293	183	467
2024	103683	5777	179	474

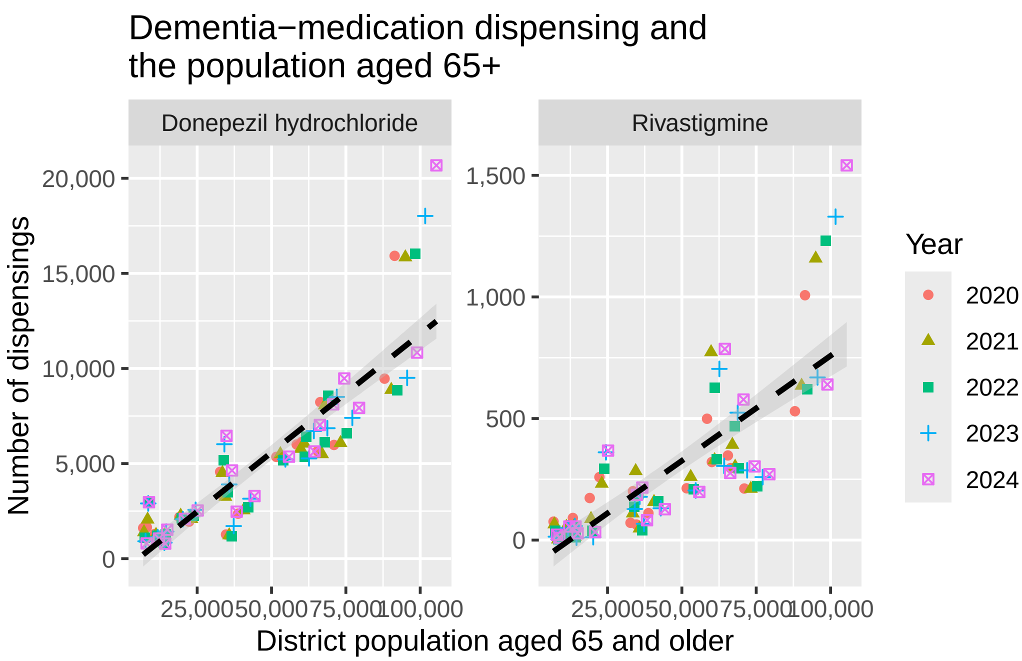
It is useful to know how frequently medications are distributed in a high concentration to a few health districts relative to an even spread across all health districts. One way to measure this is by using the proportion of dispensings going toward the health district where most of the dispensings occur. There are 19 health districts, and if the dispensings occurred equally across said districts, this metric would take the value of 1/19. In the opposite extreme, if the medication was distributed in only 1 district, the metric would take the value of 1. Below is a histogram showing how each of the medications fall on this proportion scale.



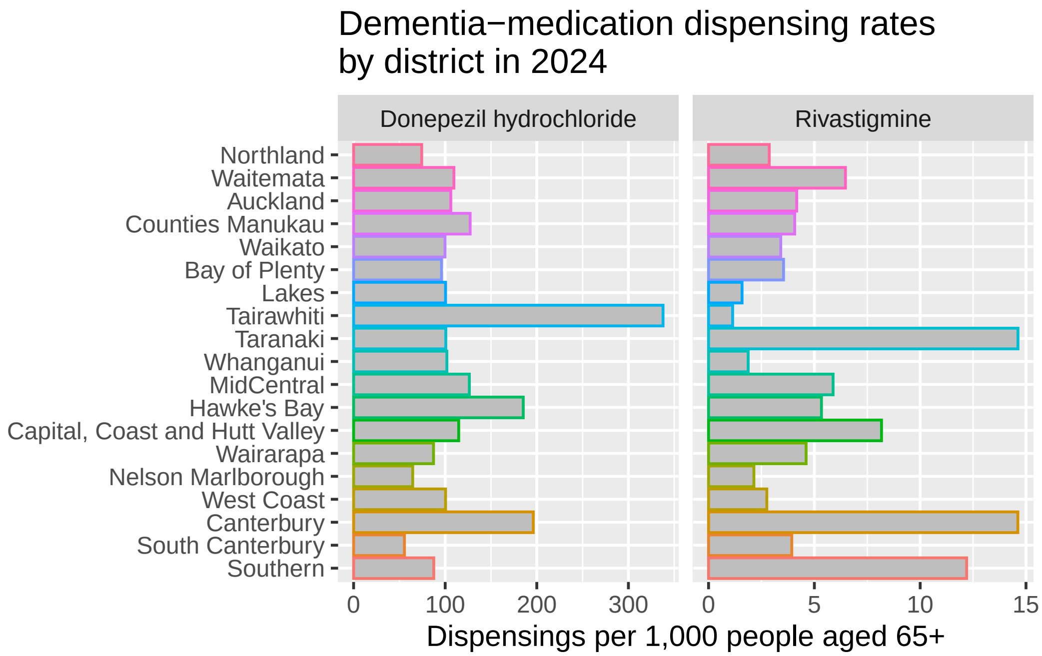
The median value is 0.1711, meaning that a typical medication will have roughly 17% of their medication go towards the district that leads in dispensing. The lowest observed proportion is

0.106283 meaning that no medication is equally distributed across all districts. There's a small group of medications that have a proportion value of 1, meaning that they are only distributed in a single district. Note that this metric doesn't speak to the number of dispensings for these medications, and are likely very niche with an estimated mean of 7.38 and max of 55. The medications with these lower dispensings may be very niche, available from retailers outside of pharmacies or not recorded in some districts, but this is speculative.

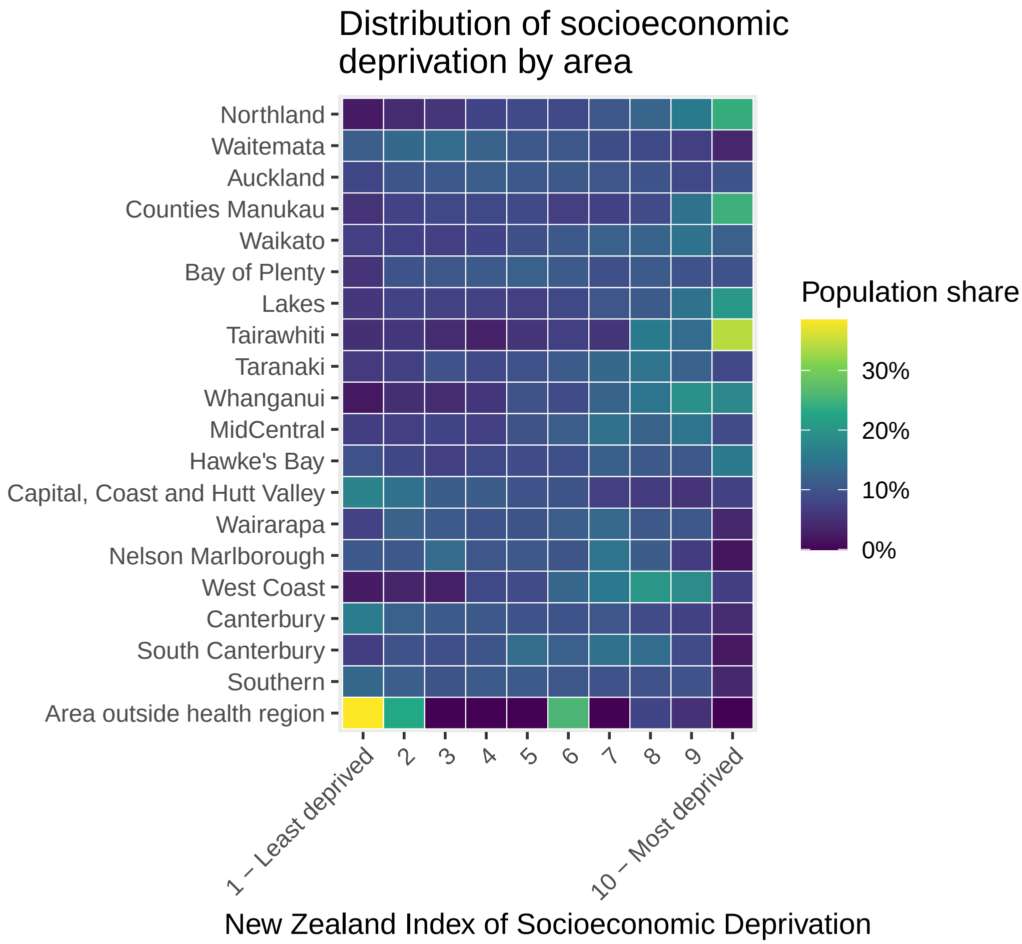
Dementia as a risk is strongly associated with old age. This makes the districts with a higher count of aged population more relevant to this analysis than total population size in the health district. The graphs below show a positive relationship between aged population count and dispensings of the two key dementia medications. This relationship is stronger for Donepezil Hydrochloride, but both show that there are some observations above the line of best fit on the higher end of the district aged population. These observations belong to the Canterbury district, and as the years increase it can be seen diverging further away from the line of best fit. Overall, this high correlation means that the older population size accounts for a large amount of the inter-district variation.



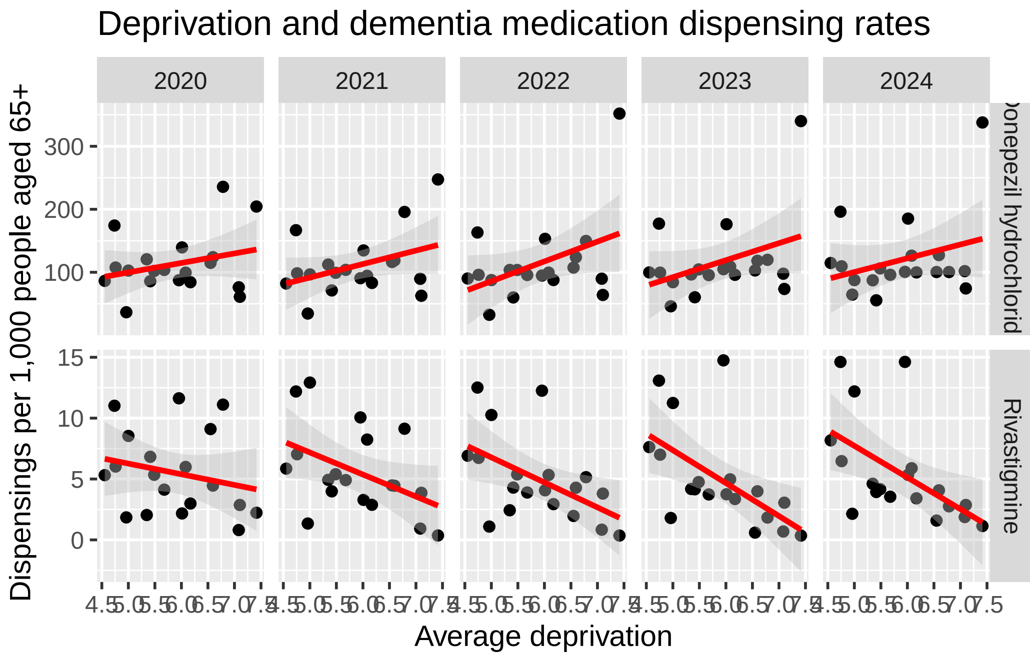
Crude rates were calculated as annual dispensings per 1,000 people aged 65 years and over for the 2024 year. The table below shows the variation in these dispensing rates across the different districts.



The first observation of note is that donepezil hydrochloride has a much higher rate than rivastigmine. Tairāwhiti has the highest overall rate when you take into account said larger rates for donepezil hydrochloride, as it is the district that has the highest rate for this medication. Canterbury and Hawke's Bay have the second and third highest rates respectively. Alternatively, Rivastigmine has the highest concentration of dispensing in Canterbury, Taranaki and Southern. Taranaki doesn't stand out for being high or low with donepezil hydrochloride, and Tairāwhiti has the lowest dispensing rate for rivastigmine. This could be related to the high proportion of the population of Tairāwhti falling higher on the latest (2023) NZ deprivation scale as seen below.



The plots below show how the average dispensing rate per aged person changes for the two dementia medications over the deprivation index. For donepezil hydrochloride, the trend in the relationship between the dispensing rate and the deprivation index increases as time increases, peaking at 2022, followed by a decreasing trend in subsequent years. Alternatively, these trends have consistently steeper downward trends as the years increase. This is a strong argument to test interaction effects between year and deprivation index when predicting the number of medication dispensings.



These results remain exploratory. There are only about 19 districts per year, the same districts are observed repeatedly, and district deprivation changes little over the study period. Confidence intervals are wide, especially for rivastigmine, and the slopes may be sensitive to influential districts such as Tairāwhiti, Canterbury or Taranaki.